Supplementary Material for Chlorophyll-a Mapping and Prediction in the Mar Menor Lagoon Using C2RCC-Processed Sentinel 2 Imagery

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Train and Test metrics

The main manuscript only contains RMSLE and R^2 metrics for the test set. This Supplementary Material provides a comprehensive breakdown of predictive performance for all combinations of models, processing methods, and aggregation windows analyzed in the study. Detailed results are presented for the R2, RMSE, and RMSLE metrics, reporting both the values obtained in the training set and in the test set for each of the water column depths evaluated.

The systematic inclusion of training metrics alongside test metrics is essential to provide a transparent view of the learning capacity of the algorithms and to allow the detection of possible signs of overfitting, thus ensuring the robustness of the generalization capacity of the selected models. All values presented correspond to the mean and standard deviation derived from the rigorous validation process using 100 random seeds, ensuring a robust statistical basis for validating the effectiveness of the proposed methodology in monitoring chlorophyll-a in the Mar Menor.

Table 1: Test RMSLE for depth 0-1 meters. The best model for each scenario is highlighted in bold.

Model	**CAT**	**ELN**	**ENS**	**KNN**	**LBM**	**LR**	**MLP**	**RF**	**SVR**	**XGB**
TOA_15x15_depth_in_0_1	0.337 ± 0.047	0.575 ± 0.045	0.362 ± 0.046	**0.331 ± 0.049**	0.385 ± 0.064	0.599 ± 0.046	0.409 ± 0.073	0.369 ± 0.052	0.426 ± 0.038	0.377 ± 0.061
C2X-Complex_rhow_15x15_depth_in_0_1	**0.337 ± 0.036**	0.469 ± 0.037	0.356 ± 0.034	0.367 ± 0.042	0.363 ± 0.043	0.508 ± 0.036	0.392 ± 0.048	0.370 ± 0.042	0.354 ± 0.033	0.353 ± 0.041
C2RCC_rhow_5x5_depth_in_0_1	**0.346 ± 0.035**	0.461 ± 0.039	0.352 ± 0.035	0.356 ± 0.044	0.375 ± 0.039	0.506 ± 0.040	0.388 ± 0.051	0.380 ± 0.045	0.368 ± 0.037	0.366 ± 0.039
C2RCC_rhown_5x5_depth_in_0_1	**0.347 ± 0.036**	0.461 ± 0.039	0.349 ± 0.034	0.356 ± 0.045	0.363 ± 0.038	0.494 ± 0.034	0.381 ± 0.047	0.378 ± 0.041	0.365 ± 0.036	0.355 ± 0.038
C2X-Complex_rhow_9x9_depth_in_0_1	**0.349 ± 0.042**	0.470 ± 0.040	0.370 ± 0.042	0.396 ± 0.054	0.378 ± 0.048	0.506 ± 0.043	0.421 ± 0.053	0.380 ± 0.048	0.360 ± 0.039	0.363 ± 0.042
C2X-Complex_rhow_5x5_depth_in_0_1	0.364 ± 0.043	0.486 ± 0.050	0.381 ± 0.048	0.389 ± 0.056	0.409 ± 0.051	0.510 ± 0.057	0.436 ± 0.050	0.399 ± 0.052	**0.356 ± 0.035**	0.396 ± 0.045
C2X-Complex_rhown_15x15_depth_in_0_1	0.359 ± 0.046	0.462 ± 0.037	0.366 ± 0.041	0.374 ± 0.048	0.383 ± 0.052	0.506 ± 0.036	0.395 ± 0.043	0.382 ± 0.047	**0.358 ± 0.033**	0.377 ± 0.046
C2X-Complex_rhown_9x9_depth_in_0_1	**0.362 ± 0.042**	0.468 ± 0.044	0.377 ± 0.046	0.401 ± 0.058	0.386 ± 0.051	0.509 ± 0.051	0.408 ± 0.058	0.396 ± 0.051	0.365 ± 0.040	0.382 ± 0.042
C2RCC_rhow_15x15_depth_in_0_1	**0.367 ± 0.048**	0.477 ± 0.034	0.375 ± 0.043	0.378 ± 0.062	0.395 ± 0.050	0.523 ± 0.033	0.416 ± 0.060	0.400 ± 0.049	0.385 ± 0.045	0.386 ± 0.048
C2RCC_rhow_9x9_depth_in_0_1	**0.369 ± 0.041**	0.482 ± 0.043	0.382 ± 0.041	0.384 ± 0.052	0.405 ± 0.045	0.504 ± 0.043	0.421 ± 0.055	0.405 ± 0.048	0.382 ± 0.042	0.391 ± 0.043

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Table 2: Train RMSLE for depth 0-1 meters.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
TOA_15x15_depth_in_0_1	0.053 ± 0.023	0.557 ± 0.014	0.226 ± 0.015	0.012 ± 0.001	0.110 ± 0.045	0.536 ± 0.027	0.328 ± 0.091	0.145 ± 0.034	0.373 ± 0.016	0.097 ± 0.056
C2X-Complex_rhow_15x15_depth_in_0_1	0.066 ± 0.039	0.450 ± 0.014	0.202 ± 0.013	0.012 ± 0.001	0.121 ± 0.044	0.465 ± 0.025	0.307 ± 0.053	0.164 ± 0.050	0.284 ± 0.012	0.088 ± 0.058
C2RCC_rhow_5x5_depth_in_0_1	0.089 ± 0.041	0.449 ± 0.011	0.208 ± 0.011	0.012 ± 0.001	0.142 ± 0.041	0.463 ± 0.012	0.315 ± 0.054	0.188 ± 0.042	0.301 ± 0.010	0.095 ± 0.046
C2RCC_rhown_5x5_depth_in_0_1	0.084 ± 0.042	0.450 ± 0.011	0.207 ± 0.012	0.012 ± 0.001	0.142 ± 0.041	0.459 ± 0.016	0.304 ± 0.046	0.177 ± 0.046	0.300 ± 0.010	0.108 ± 0.045
C2X-Complex_rhow_9x9_depth_in_0_1	0.085 ± 0.045	0.449 ± 0.014	0.211 ± 0.013	0.012 ± 0.001	0.141 ± 0.049	0.467 ± 0.017	0.324 ± 0.063	0.176 ± 0.044	0.280 ± 0.011	0.103 ± 0.050
C2X-Complex_rhow_5x5_depth_in_0_1	0.069 ± 0.040	0.470 ± 0.020	0.217 ± 0.014	0.012 ± 0.001	0.147 ± 0.046	0.491 ± 0.016	0.311 ± 0.059	0.179 ± 0.054	0.272 ± 0.010	0.122 ± 0.074
C2X-Complex_rhown_15x15_depth_in_0_1	0.074 ± 0.044	0.449 ± 0.014	0.217 ± 0.013	0.012 ± 0.001	0.147 ± 0.039	0.472 ± 0.024	0.320 ± 0.046	0.189 ± 0.044	0.286 ± 0.010	0.140 ± 0.059
C2X-Complex_rhown_9x9_depth_in_0_1	0.073 ± 0.046	0.452 ± 0.014	0.220 ± 0.013	0.012 ± 0.001	0.148 ± 0.043	0.477 ± 0.019	0.332 ± 0.058	0.194 ± 0.053	0.286 ± 0.012	0.119 ± 0.059
C2RCC_rhow_15x15_depth_in_0_1	0.104 ± 0.047	0.466 ± 0.011	0.237 ± 0.014	0.012 ± 0.001	0.165 ± 0.039	0.494 ± 0.015	0.340 ± 0.068	0.201 ± 0.050	0.331 ± 0.011	0.149 ± 0.052
C2RCC_rhow_9x9_depth_in_0_1	0.099 ± 0.051	0.467 ± 0.012	0.231 ± 0.015	0.012 ± 0.001	0.156 ± 0.046	0.470 ± 0.015	0.352 ± 0.065	0.194 ± 0.051	0.323 ± 0.011	0.129 ± 0.058

Table 3: Test R^2 for depth 0-1 meters. The best model for each scenario is highlighted in bold.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
C2RCC_rhow_5x5_depth_in_0_1	**0.767 ± 0.082**	0.594 ± 0.089	0.768 ± 0.072	0.756 ± 0.099	0.707 ± 0.118	0.497 ± 0.168	0.743 ± 0.138	0.692 ± 0.131	0.706 ± 0.086	0.734 ± 0.100
C2RCC_rhown_5x5_depth_in_0_1	0.763 ± 0.083	0.598 ± 0.090	**0.767 ± 0.070**	0.758 ± 0.093	0.719 ± 0.123	0.504 ± 0.146	0.751 ± 0.113	0.690 ± 0.131	0.701 ± 0.078	0.742 ± 0.106
TOA_15x15_depth_in_0_1	0.743 ± 0.117	0.479 ± 0.143	0.725 ± 0.092	**0.764 ± 0.131**	0.626 ± 0.201	0.437 ± 0.166	0.659 ± 0.209	0.692 ± 0.156	0.411 ± 0.091	0.618 ± 0.195
C2X-Complex_rhow_9x9_depth_in_0_1	**0.763 ± 0.082**	0.577 ± 0.157	0.736 ± 0.083	0.670 ± 0.122	0.708 ± 0.122	0.485 ± 0.339	0.656 ± 0.139	0.706 ± 0.111	0.699 ± 0.098	0.732 ± 0.102
C2X-Complex_rhow_15x15_depth_in_0_1	**0.755 ± 0.098**	0.586 ± 0.120	0.729 ± 0.086	0.711 ± 0.110	0.662 ± 0.143	0.471 ± 0.284	0.704 ± 0.112	0.662 ± 0.140	0.698 ± 0.111	0.696 ± 0.118
C2X-Complex_rhow_5x5_depth_in_0_1	**0.737 ± 0.089**	0.498 ± 0.153	0.709 ± 0.089	0.666 ± 0.126	0.639 ± 0.143	0.380 ± 0.424	0.647 ± 0.131	0.674 ± 0.125	0.693 ± 0.107	0.640 ± 0.126
C2RCC_rhow_15x15_depth_in_0_1	0.730 ± 0.098	0.540 ± 0.115	0.723 ± 0.089	**0.733 ± 0.148**	0.669 ± 0.132	0.477 ± 0.215	0.682 ± 0.147	0.648 ± 0.137	0.677 ± 0.097	0.679 ± 0.131
C2X-Complex_rhown_9x9_depth_in_0_1	**0.733 ± 0.092**	0.583 ± 0.132	0.716 ± 0.088	0.671 ± 0.123	0.667 ± 0.128	0.510 ± 0.244	0.673 ± 0.141	0.651 ± 0.133	0.689 ± 0.096	0.674 ± 0.109
C2RCC_rhow_9x9_depth_in_0_1	**0.726 ± 0.098**	0.595 ± 0.096	0.721 ± 0.086	0.724 ± 0.127	0.658 ± 0.136	0.522 ± 0.168	0.678 ± 0.158	0.652 ± 0.131	0.691 ± 0.092	0.691 ± 0.119
C2X-Complex_rhown_15x15_depth_in_0_1	**0.709 ± 0.115**	0.590 ± 0.111	0.705 ± 0.100	0.702 ± 0.125	0.639 ± 0.178	0.535 ± 0.131	0.686 ± 0.113	0.636 ± 0.160	0.691 ± 0.103	0.640 ± 0.138

Table 4: Train R^2 for depth 0-1 meters.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
C2RCC_rhow_5x5_depth_in_0_1	0.997 ± 0.003	0.656 ± 0.022	0.948 ± 0.008	1.000 ± 0.000	0.991 ± 0.006	0.673 ± 0.078	0.855 ± 0.116	0.952 ± 0.024	0.811 ± 0.040	0.993 ± 0.009
C2RCC_rhown_5x5_depth_in_0_1	0.997 ± 0.003	0.657 ± 0.022	0.947 ± 0.010	1.000 ± 0.000	0.991 ± 0.006	0.650 ± 0.076	0.870 ± 0.082	0.954 ± 0.026	0.800 ± 0.039	0.991 ± 0.011
TOA_15x15_depth_in_0_1	0.999 ± 0.001	0.577 ± 0.027	0.936 ± 0.014	1.000 ± 0.000	0.991 ± 0.009	0.659 ± 0.053	0.802 ± 0.195	0.961 ± 0.029	0.567 ± 0.059	0.972 ± 0.074
C2X-Complex_rhow_9x9_depth_in_0_1	0.997 ± 0.004	0.676 ± 0.041	0.944 ± 0.010	1.000 ± 0.000	0.990 ± 0.006	0.673 ± 0.067	0.816 ± 0.127	0.957 ± 0.028	0.787 ± 0.031	0.990 ± 0.032
C2X-Complex_rhow_15x15_depth_in_0_1	0.998 ± 0.003	0.674 ± 0.045	0.945 ± 0.012	1.000 ± 0.000	0.993 ± 0.005	0.664 ± 0.089	0.834 ± 0.093	0.959 ± 0.028	0.785 ± 0.027	0.976 ± 0.078
C2X-Complex_rhow_5x5_depth_in_0_1	0.998 ± 0.002	0.604 ± 0.074	0.934 ± 0.018	1.000 ± 0.000	0.990 ± 0.006	0.567 ± 0.054	0.817 ± 0.134	0.960 ± 0.036	0.789 ± 0.030	0.937 ± 0.144
C2RCC_rhow_15x15_depth_in_0_1	0.996 ± 0.004	0.613 ± 0.028	0.931 ± 0.014	1.000 ± 0.000	0.986 ± 0.008	0.623 ± 0.084	0.804 ± 0.138	0.944 ± 0.033	0.775 ± 0.041	0.962 ± 0.084
C2X-Complex_rhown_9x9_depth_in_0_1	0.997 ± 0.004	0.661 ± 0.034	0.938 ± 0.012	1.000 ± 0.000	0.989 ± 0.007	0.646 ± 0.073	0.822 ± 0.108	0.937 ± 0.038	0.784 ± 0.034	0.969 ± 0.085
C2RCC_rhow_9x9_depth_in_0_1	0.996 ± 0.006	0.656 ± 0.027	0.935 ± 0.014	1.000 ± 0.000	0.987 ± 0.008	0.650 ± 0.079	0.793 ± 0.144	0.950 ± 0.029	0.780 ± 0.048	0.976 ± 0.061
C2X-Complex_rhown_15x15_depth_in_0_1	0.998 ± 0.004	0.659 ± 0.035	0.935 ± 0.013	1.000 ± 0.000	0.990 ± 0.006	0.649 ± 0.083	0.834 ± 0.080	0.936 ± 0.032	0.784 ± 0.025	0.944 ± 0.117

Table 5: Test RMSE for depth 0-1 meters. The best model for each scenario is highlighted in bold.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
TOA_15x15_depth_in_0_1	1.881 ± 0.525	2.702 ± 0.348	1.968 ± 0.432	**1.777 ± 0.508**	2.240 ± 0.642	2.802 ± 0.354	2.126 ± 0.702	2.052 ± 0.586	2.916 ± 0.444	2.271 ± 0.634
C2RCC_rhow_5x5_depth_in_0_1	**1.830 ± 0.307**	2.439 ± 0.245	1.832 ± 0.274	1.858 ± 0.334	2.040 ± 0.410	2.692 ± 0.409	1.890 ± 0.423	2.083 ± 0.407	2.070 ± 0.314	1.948 ± 0.337
C2RCC_rhown_5x5_depth_in_0_1	1.843 ± 0.309	2.425 ± 0.243	**1.836 ± 0.272**	1.856 ± 0.344	1.988 ± 0.420	2.683 ± 0.370	1.877 ± 0.391	2.090 ± 0.376	2.091 ± 0.290	1.912 ± 0.354
C2X-Complex_rhow_9x9_depth_in_0_1	**1.840 ± 0.315**	2.469 ± 0.381	1.957 ± 0.309	2.177 ± 0.397	2.029 ± 0.370	2.672 ± 0.668	2.219 ± 0.442	2.055 ± 0.400	2.095 ± 0.407	1.953 ± 0.336
C2X-Complex_rhow_15x15_depth_in_0_1	**1.860 ± 0.393**	2.440 ± 0.321	1.971 ± 0.348	2.033 ± 0.423	2.168 ± 0.452	2.717 ± 0.636	2.052 ± 0.395	2.182 ± 0.457	2.081 ± 0.462	2.064 ± 0.379
C2RCC_rhow_15x15_depth_in_0_1	1.957 ± 0.341	2.575 ± 0.262	1.991 ± 0.290	**1.903 ± 0.473**	2.159 ± 0.410	2.710 ± 0.424	2.101 ± 0.450	2.227 ± 0.396	2.156 ± 0.308	2.126 ± 0.414
C2X-Complex_rhow_5x5_depth_in_0_1	**1.949 ± 0.354**	2.706 ± 0.442	2.061 ± 0.358	2.189 ± 0.441	2.267 ± 0.425	2.933 ± 0.707	2.260 ± 0.452	2.163 ± 0.438	2.107 ± 0.418	2.278 ± 0.440
C2X-Complex_rhown_9x9_depth_in_0_1	**1.958 ± 0.333**	2.458 ± 0.351	2.030 ± 0.311	2.173 ± 0.403	2.173 ± 0.400	2.631 ± 0.506	2.159 ± 0.423	2.236 ± 0.411	2.128 ± 0.378	2.163 ± 0.349
C2RCC_rhow_9x9_depth_in_0_1	1.982 ± 0.327	2.432 ± 0.254	2.010 ± 0.287	**1.964 ± 0.407**	2.208 ± 0.412	2.624 ± 0.381	2.128 ± 0.506	2.233 ± 0.402	2.118 ± 0.308	2.103 ± 0.400
C2X-Complex_rhown_15x15_depth_in_0_1	**2.024 ± 0.406**	2.428 ± 0.280	2.054 ± 0.344	2.046 ± 0.391	2.233 ± 0.520	2.587 ± 0.341	2.117 ± 0.390	2.259 ± 0.487	2.106 ± 0.396	2.254 ± 0.419

Table 6: Train RMSE for depth 0-1 meters.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
TOA_15x15_depth_in_0_1	0.108 ± 0.057	2.488 ± 0.086	0.964 ± 0.102	0.014 ± 0.001	0.316 ± 0.173	2.230 ± 0.152	1.533 ± 0.736	0.719 ± 0.244	2.513 ± 0.168	0.418 ± 0.497
C2RCC_rhow_5x5_depth_in_0_1	0.184 ± 0.099	2.240 ± 0.063	0.868 ± 0.067	0.014 ± 0.001	0.347 ± 0.120	2.169 ± 0.258	1.400 ± 0.394	0.811 ± 0.216	1.653 ± 0.148	0.255 ± 0.176
C2RCC_rhown_5x5_depth_in_0_1	0.174 ± 0.102	2.237 ± 0.060	0.876 ± 0.073	0.014 ± 0.001	0.345 ± 0.123	2.246 ± 0.237	1.334 ± 0.336	0.786 ± 0.231	1.698 ± 0.145	0.300 ± 0.182
C2X-Complex_rhow_9x9_depth_in_0_1	0.180 ± 0.111	2.172 ± 0.124	0.903 ± 0.078	0.014 ± 0.001	0.348 ± 0.144	2.174 ± 0.220	1.577 ± 0.441	0.753 ± 0.242	1.759 ± 0.129	0.289 ± 0.258
C2X-Complex_rhow_15x15_depth_in_0_1	0.134 ± 0.091	2.178 ± 0.141	0.890 ± 0.092	0.014 ± 0.001	0.291 ± 0.128	2.195 ± 0.293	1.510 ± 0.375	0.732 ± 0.264	1.771 ± 0.126	0.335 ± 0.495
C2RCC_rhow_15x15_depth_in_0_1	0.221 ± 0.117	2.376 ± 0.078	1.001 ± 0.101	0.014 ± 0.001	0.423 ± 0.132	2.330 ± 0.275	1.617 ± 0.499	0.863 ± 0.282	1.804 ± 0.148	0.561 ± 0.479
C2X-Complex_rhow_5x5_depth_in_0_1	0.142 ± 0.094	2.395 ± 0.226	0.970 ± 0.120	0.014 ± 0.001	0.351 ± 0.137	2.510 ± 0.176	1.557 ± 0.496	0.706 ± 0.293	1.750 ± 0.125	0.557 ± 0.775
C2X-Complex_rhown_9x9_depth_in_0_1	0.153 ± 0.115	2.223 ± 0.110	0.949 ± 0.092	0.014 ± 0.001	0.374 ± 0.135	2.261 ± 0.245	1.559 ± 0.403	0.913 ± 0.294	1.771 ± 0.139	0.447 ± 0.506
C2RCC_rhow_9x9_depth_in_0_1	0.211 ± 0.138	2.238 ± 0.081	0.966 ± 0.102	0.014 ± 0.001	0.403 ± 0.148	2.245 ± 0.260	1.673 ± 0.471	0.812 ± 0.258	1.783 ± 0.183	0.447 ± 0.388
C2X-Complex_rhown_15x15_depth_in_0_1	0.155 ± 0.108	2.230 ± 0.113	0.968 ± 0.100	0.014 ± 0.001	0.365 ± 0.120	2.247 ± 0.268	1.516 ± 0.358	0.930 ± 0.261	1.772 ± 0.115	0.641 ± 0.645

Table 7: Test RMSLE for depth 1-2 meters. The best model for each scenario is highlighted in bold.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
C2X-Complex_rhow_15x15_depth_in_1_2	**0.365 ± 0.048**	0.475 ± 0.051	0.385 ± 0.048	0.392 ± 0.055	0.394 ± 0.053	0.511 ± 0.052	0.429 ± 0.049	0.400 ± 0.057	0.375 ± 0.042	0.380 ± 0.050
C2X-Complex_rhow_9x9_depth_in_1_2	**0.382 ± 0.047**	0.476 ± 0.053	0.398 ± 0.051	0.419 ± 0.059	0.393 ± 0.053	0.509 ± 0.053	0.454 ± 0.065	0.402 ± 0.048	0.387 ± 0.047	0.389 ± 0.048
C2X-Complex_rhown_9x9_depth_in_1_2	**0.392 ± 0.047**	0.474 ± 0.053	0.403 ± 0.052	0.425 ± 0.064	0.412 ± 0.048	0.511 ± 0.050	0.442 ± 0.064	0.418 ± 0.053	0.395 ± 0.049	0.406 ± 0.043
C2X-Complex_rhow_5x5_depth_in_1_2	0.395 ± 0.047	0.493 ± 0.055	0.409 ± 0.052	0.418 ± 0.054	0.426 ± 0.050	0.520 ± 0.058	0.463 ± 0.060	0.421 ± 0.053	**0.393 ± 0.038**	0.412 ± 0.048
C2X-Complex_rhow_3x3_depth_in_1_2	**0.394 ± 0.042**	0.493 ± 0.053	0.412 ± 0.049	0.420 ± 0.051	0.422 ± 0.051	0.517 ± 0.057	0.466 ± 0.050	0.428 ± 0.052	0.409 ± 0.035	0.416 ± 0.046
C2X-Complex_rhown_3x3_depth_in_1_2	**0.395 ± 0.045**	0.497 ± 0.055	0.412 ± 0.051	0.426 ± 0.051	0.419 ± 0.058	0.524 ± 0.052	0.454 ± 0.055	0.425 ± 0.054	0.413 ± 0.038	0.410 ± 0.052
C2RCC_rhown_3x3_depth_in_1_2	**0.400 ± 0.041**	0.486 ± 0.048	0.405 ± 0.043	0.404 ± 0.046	0.425 ± 0.046	0.538 ± 0.052	0.423 ± 0.049	0.436 ± 0.050	0.408 ± 0.042	0.419 ± 0.044
C2X-Complex_rhown_5x5_depth_in_1_2	0.402 ± 0.045	0.489 ± 0.055	0.413 ± 0.051	0.424 ± 0.053	0.432 ± 0.051	0.524 ± 0.050	0.450 ± 0.047	0.429 ± 0.053	**0.401 ± 0.042**	0.421 ± 0.048
C2X_rhow_5x5_depth_in_1_2	0.407 ± 0.041	0.499 ± 0.049	0.418 ± 0.042	0.422 ± 0.040	0.427 ± 0.052	0.527 ± 0.054	0.462 ± 0.056	0.429 ± 0.048	**0.406 ± 0.035**	0.424 ± 0.046
C2X_rhow_3x3_depth_in_1_2	0.422 ± 0.041	0.489 ± 0.049	0.427 ± 0.044	0.433 ± 0.045	0.442 ± 0.050	0.519 ± 0.049	0.471 ± 0.052	0.443 ± 0.047	**0.421 ± 0.036**	0.440 ± 0.047

Table 8: Train RMSLE for depth 1-2 meters.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
C2X-Complex_rhow_15x15_depth_in_1_2	0.091 ± 0.047	0.448 ± 0.013	0.220 ± 0.016	0.009 ± 0.002	0.139 ± 0.047	0.460 ± 0.020	0.339 ± 0.056	0.173 ± 0.057	0.299 ± 0.012	0.104 ± 0.063
C2X-Complex_rhow_9x9_depth_in_1_2	0.126 ± 0.067	0.455 ± 0.015	0.238 ± 0.019	0.009 ± 0.002	0.157 ± 0.046	0.467 ± 0.013	0.382 ± 0.065	0.196 ± 0.054	0.301 ± 0.013	0.121 ± 0.065
C2X-Complex_rhown_9x9_depth_in_1_2	0.089 ± 0.056	0.457 ± 0.014	0.237 ± 0.016	0.009 ± 0.002	0.169 ± 0.033	0.467 ± 0.013	0.364 ± 0.067	0.202 ± 0.055	0.306 ± 0.015	0.140 ± 0.063
C2X-Complex_rhow_5x5_depth_in_1_2	0.092 ± 0.057	0.470 ± 0.017	0.240 ± 0.017	0.009 ± 0.002	0.155 ± 0.054	0.491 ± 0.018	0.383 ± 0.065	0.204 ± 0.060	0.296 ± 0.018	0.095 ± 0.068
C2X-Complex_rhow_3x3_depth_in_1_2	0.141 ± 0.061	0.463 ± 0.015	0.248 ± 0.019	0.009 ± 0.002	0.162 ± 0.048	0.478 ± 0.014	0.386 ± 0.057	0.224 ± 0.069	0.323 ± 0.018	0.147 ± 0.071
C2X-Complex_rhown_3x3_depth_in_1_2	0.071 ± 0.046	0.471 ± 0.015	0.242 ± 0.015	0.009 ± 0.002	0.154 ± 0.052	0.491 ± 0.016	0.378 ± 0.057	0.208 ± 0.067	0.339 ± 0.016	0.118 ± 0.075
C2RCC_rhown_3x3_depth_in_1_2	0.125 ± 0.055	0.467 ± 0.010	0.246 ± 0.015	0.009 ± 0.002	0.184 ± 0.050	0.488 ± 0.017	0.360 ± 0.041	0.237 ± 0.060	0.332 ± 0.015	0.155 ± 0.067
C2X-Complex_rhown_5x5_depth_in_1_2	0.092 ± 0.059	0.472 ± 0.016	0.244 ± 0.017	0.009 ± 0.002	0.160 ± 0.051	0.492 ± 0.020	0.371 ± 0.048	0.215 ± 0.063	0.310 ± 0.021	0.119 ± 0.077
C2X_rhow_5x5_depth_in_1_2	0.154 ± 0.060	0.454 ± 0.014	0.254 ± 0.016	0.027 ± 0.004	0.187 ± 0.035	0.455 ± 0.029	0.406 ± 0.067	0.226 ± 0.058	0.328 ± 0.012	0.192 ± 0.059
C2X_rhow_3x3_depth_in_1_2	0.170 ± 0.062	0.446 ± 0.013	0.264 ± 0.016	0.027 ± 0.004	0.194 ± 0.041	0.466 ± 0.022	0.416 ± 0.056	0.258 ± 0.053	0.346 ± 0.013	0.205 ± 0.059

Table 9: Test R^2 for depth 1-2 meters. The best model for each scenario is highlighted in bold.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
C2X-Complex_rhow_15x15_depth_in_1_2	**0.759 ± 0.097**	0.596 ± 0.174	0.729 ± 0.111	0.714 ± 0.142	0.670 ± 0.190	0.386 ± 0.483	0.684 ± 0.128	0.656 ± 0.181	0.728 ± 0.104	0.715 ± 0.123
C2X-Complex_rhow_9x9_depth_in_1_2	**0.756 ± 0.104**	0.567 ± 0.207	0.723 ± 0.116	0.673 ± 0.148	0.715 ± 0.148	0.420 ± 0.391	0.613 ± 0.202	0.705 ± 0.125	0.740 ± 0.085	0.728 ± 0.126
C2X-Complex_rhow_5x5_depth_in_1_2	**0.748 ± 0.096**	0.503 ± 0.240	0.711 ± 0.130	0.664 ± 0.173	0.669 ± 0.166	0.394 ± 0.439	0.625 ± 0.173	0.672 ± 0.159	0.747 ± 0.085	0.696 ± 0.126
C2RCC_rhown_3x3_depth_in_1_2	**0.743 ± 0.115**	0.564 ± 0.146	0.730 ± 0.113	0.732 ± 0.129	0.666 ± 0.181	0.417 ± 0.292	0.727 ± 0.131	0.649 ± 0.191	0.722 ± 0.115	0.681 ± 0.158
C2X-Complex_rhow_3x3_depth_in_1_2	**0.742 ± 0.096**	0.510 ± 0.296	0.690 ± 0.144	0.662 ± 0.147	0.646 ± 0.187	0.315 ± 0.931	0.639 ± 0.173	0.635 ± 0.159	0.706 ± 0.093	0.663 ± 0.135
C2X-Complex_rhown_5x5_depth_in_1_2	**0.732 ± 0.102**	0.504 ± 0.277	0.704 ± 0.123	0.659 ± 0.158	0.661 ± 0.160	0.348 ± 0.437	0.653 ± 0.177	0.666 ± 0.150	0.717 ± 0.100	0.673 ± 0.129
C2X-Complex_rhown_9x9_depth_in_1_2	0.724 ± 0.117	0.596 ± 0.165	0.707 ± 0.122	0.663 ± 0.167	0.645 ± 0.184	0.486 ± 0.312	0.667 ± 0.198	0.632 ± 0.172	**0.728 ± 0.096**	0.671 ± 0.140
C2X-Complex_rhown_3x3_depth_in_1_2	**0.725 ± 0.118**	0.482 ± 0.273	0.682 ± 0.138	0.634 ± 0.170	0.655 ± 0.161	0.324 ± 0.469	0.644 ± 0.196	0.636 ± 0.161	0.671 ± 0.117	0.670 ± 0.136
C2X_rhow_5x5_depth_in_1_2	**0.675 ± 0.111**	0.536 ± 0.151	0.661 ± 0.098	0.647 ± 0.110	0.619 ± 0.146	0.503 ± 0.184	0.600 ± 0.122	0.636 ± 0.128	0.648 ± 0.087	0.608 ± 0.129
C2X_rhow_3x3_depth_in_1_2	**0.638 ± 0.140**	0.535 ± 0.168	0.630 ± 0.104	0.619 ± 0.102	0.600 ± 0.156	0.497 ± 0.190	0.580 ± 0.139	0.605 ± 0.124	0.595 ± 0.085	0.575 ± 0.141

Table 10: Train R^2 for depth 1-2 meters.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
C2X-Complex_rhow_15x15_depth_in_1_2	0.996 ± 0.004	0.705 ± 0.035	0.947 ± 0.011	1.000 ± 0.000	0.990 ± 0.007	0.694 ± 0.070	0.807 ± 0.087	0.958 ± 0.027	0.839 ± 0.026	0.989 ± 0.013
C2X-Complex_rhow_9x9_depth_in_1_2	0.991 ± 0.011	0.694 ± 0.034	0.941 ± 0.011	1.000 ± 0.000	0.988 ± 0.007	0.690 ± 0.053	0.751 ± 0.132	0.956 ± 0.026	0.848 ± 0.027	0.987 ± 0.015
C2X-Complex_rhow_5x5_depth_in_1_2	0.995 ± 0.007	0.646 ± 0.060	0.940 ± 0.013	1.000 ± 0.000	0.988 ± 0.008	0.621 ± 0.058	0.772 ± 0.119	0.955 ± 0.044	0.866 ± 0.030	0.979 ± 0.071
C2RCC_rhown_3x3_depth_in_1_2	0.992 ± 0.008	0.636 ± 0.030	0.930 ± 0.015	1.000 ± 0.000	0.981 ± 0.010	0.629 ± 0.077	0.821 ± 0.075	0.932 ± 0.049	0.812 ± 0.031	0.948 ± 0.111
C2X-Complex_rhow_3x3_depth_in_1_2	0.990 ± 0.009	0.676 ± 0.052	0.931 ± 0.016	1.000 ± 0.000	0.987 ± 0.009	0.646 ± 0.064	0.775 ± 0.097	0.934 ± 0.060	0.824 ± 0.037	0.948 ± 0.107
C2X-Complex_rhown_5x5_depth_in_1_2	0.995 ± 0.007	0.639 ± 0.046	0.936 ± 0.015	1.000 ± 0.000	0.987 ± 0.008	0.628 ± 0.067	0.798 ± 0.090	0.950 ± 0.046	0.836 ± 0.044	0.960 ± 0.106
C2X-Complex_rhown_9x9_depth_in_1_2	0.996 ± 0.006	0.686 ± 0.030	0.941 ± 0.011	1.000 ± 0.000	0.986 ± 0.006	0.694 ± 0.062	0.805 ± 0.104	0.941 ± 0.033	0.847 ± 0.028	0.976 ± 0.056
C2X-Complex_rhown_3x3_depth_in_1_2	0.997 ± 0.005	0.647 ± 0.046	0.931 ± 0.015	1.000 ± 0.000	0.988 ± 0.007	0.619 ± 0.053	0.789 ± 0.094	0.942 ± 0.059	0.791 ± 0.042	0.951 ± 0.115
C2X_rhow_5x5_depth_in_1_2	0.988 ± 0.010	0.700 ± 0.030	0.925 ± 0.015	1.000 ± 0.000	0.982 ± 0.008	0.735 ± 0.062	0.737 ± 0.108	0.934 ± 0.042	0.797 ± 0.035	0.908 ± 0.145
C2X_rhow_3x3_depth_in_1_2	0.984 ± 0.012	0.704 ± 0.026	0.912 ± 0.019	1.000 ± 0.000	0.981 ± 0.008	0.697 ± 0.055	0.712 ± 0.103	0.906 ± 0.046	0.748 ± 0.042	0.893 ± 0.156

Table 11: Test RMSE for depth 1-2 meters. The best model for each scenario is highlighted in bold.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
C2X-Complex_rhow_15x15_depth_in_1_2	**1.874 ± 0.313**	2.421 ± 0.365	1.989 ± 0.315	2.021 ± 0.396	2.148 ± 0.458	2.875 ± 0.886	2.153 ± 0.374	2.217 ± 0.434	2.002 ± 0.351	2.032 ± 0.362
C2X-Complex_rhow_9x9_depth_in_1_2	**1.883 ± 0.291**	2.499 ± 0.434	2.012 ± 0.315	2.177 ± 0.385	2.016 ± 0.421	2.826 ± 0.792	2.358 ± 0.498	2.076 ± 0.355	1.967 ± 0.303	1.982 ± 0.322
C2X-Complex_rhow_5x5_depth_in_1_2	**1.918 ± 0.278**	2.689 ± 0.535	2.051 ± 0.366	2.196 ± 0.427	2.176 ± 0.455	2.917 ± 0.823	2.329 ± 0.424	2.180 ± 0.445	1.934 ± 0.269	2.103 ± 0.387
C2RCC_rhown_3x3_depth_in_1_2	**1.938 ± 0.360**	2.549 ± 0.370	1.992 ± 0.336	1.972 ± 0.392	2.189 ± 0.452	2.905 ± 0.591	1.983 ± 0.353	2.240 ± 0.460	2.021 ± 0.344	2.153 ± 0.440
C2X-Complex_rhow_3x3_depth_in_1_2	**1.943 ± 0.249**	2.646 ± 0.625	2.127 ± 0.385	2.220 ± 0.379	2.250 ± 0.471	2.976 ± 1.306	2.286 ± 0.415	2.305 ± 0.425	2.089 ± 0.264	2.224 ± 0.386
C2X-Complex_rhown_5x5_depth_in_1_2	**1.979 ± 0.264**	2.668 ± 0.567	2.080 ± 0.343	2.218 ± 0.413	2.205 ± 0.428	3.021 ± 0.829	2.230 ± 0.377	2.200 ± 0.432	2.043 ± 0.263	2.189 ± 0.389
C2X-Complex_rhown_3x3_depth_in_1_2	**1.996 ± 0.274**	2.735 ± 0.620	2.155 ± 0.353	2.294 ± 0.381	2.223 ± 0.442	3.069 ± 0.814	2.252 ± 0.454	2.306 ± 0.438	2.206 ± 0.249	2.199 ± 0.417
C2X-Complex_rhown_9x9_depth_in_1_2	**2.006 ± 0.293**	2.432 ± 0.332	2.070 ± 0.299	2.203 ± 0.424	2.251 ± 0.445	2.686 ± 0.585	2.178 ± 0.449	2.308 ± 0.412	2.004 ± 0.273	2.193 ± 0.352
C2X_rhow_5x5_depth_in_1_2	**2.196 ± 0.362**	2.620 ± 0.345	2.252 ± 0.298	2.294 ± 0.339	2.363 ± 0.389	2.704 ± 0.419	2.445 ± 0.366	2.322 ± 0.339	2.303 ± 0.309	2.415 ± 0.415
C2X_rhow_3x3_depth_in_1_2	**2.309 ± 0.334**	2.619 ± 0.392	2.353 ± 0.298	2.391 ± 0.303	2.422 ± 0.390	2.722 ± 0.423	2.502 ± 0.396	2.424 ± 0.316	2.481 ± 0.334	2.516 ± 0.428

Table 12: Train RMSE for depth 1-2 meters.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
C2X-Complex_rhow_15x15_depth_in_1_2	0.207 ± 0.123	2.121 ± 0.111	0.900 ± 0.081	0.010 ± 0.002	0.366 ± 0.146	2.151 ± 0.255	1.676 ± 0.375	0.754 ± 0.269	1.564 ± 0.117	0.323 ± 0.253
C2X-Complex_rhow_9x9_depth_in_1_2	0.314 ± 0.204	2.160 ± 0.110	0.949 ± 0.087	0.010 ± 0.002	0.405 ± 0.139	2.172 ± 0.185	1.901 ± 0.456	0.781 ± 0.241	1.519 ± 0.124	0.366 ± 0.255
C2X-Complex_rhow_5x5_depth_in_1_2	0.220 ± 0.162	2.322 ± 0.198	0.954 ± 0.096	0.010 ± 0.002	0.401 ± 0.166	2.403 ± 0.199	1.814 ± 0.452	0.761 ± 0.332	1.423 ± 0.154	0.317 ± 0.466
C2RCC_rhown_3x3_depth_in_1_2	0.304 ± 0.162	2.360 ± 0.099	1.028 ± 0.106	0.010 ± 0.002	0.505 ± 0.168	2.370 ± 0.264	1.628 ± 0.289	0.963 ± 0.326	1.690 ± 0.138	0.638 ± 0.621
C2X-Complex_rhow_3x3_depth_in_1_2	0.355 ± 0.177	2.219 ± 0.164	1.022 ± 0.114	0.010 ± 0.002	0.426 ± 0.152	2.318 ± 0.222	1.818 ± 0.376	0.913 ± 0.418	1.634 ± 0.163	0.610 ± 0.657
C2X-Complex_rhown_5x5_depth_in_1_2	0.219 ± 0.164	2.347 ± 0.142	0.983 ± 0.102	0.010 ± 0.002	0.416 ± 0.156	2.376 ± 0.230	1.723 ± 0.362	0.808 ± 0.339	1.571 ± 0.196	0.471 ± 0.625
C2X-Complex_rhown_3x3_depth_in_1_2	0.163 ± 0.124	2.320 ± 0.140	1.018 ± 0.095	0.010 ± 0.002	0.393 ± 0.152	2.410 ± 0.173	1.759 ± 0.374	0.843 ± 0.404	1.778 ± 0.171	0.510 ± 0.690
C2X-Complex_rhown_9x9_depth_in_1_2	0.208 ± 0.154	2.189 ± 0.097	0.942 ± 0.083	0.010 ± 0.002	0.452 ± 0.110	2.153 ± 0.216	1.673 ± 0.412	0.904 ± 0.276	1.523 ± 0.128	0.461 ± 0.375
C2X_rhow_5x5_depth_in_1_2	0.389 ± 0.184	2.140 ± 0.106	1.064 ± 0.109	0.040 ± 0.006	0.510 ± 0.128	2.003 ± 0.234	1.972 ± 0.390	0.952 ± 0.328	1.759 ± 0.151	0.910 ± 0.763
C2X_rhow_3x3_depth_in_1_2	0.446 ± 0.201	2.128 ± 0.094	1.154 ± 0.119	0.040 ± 0.006	0.522 ± 0.136	2.145 ± 0.201	2.067 ± 0.353	1.157 ± 0.316	1.959 ± 0.176	1.006 ± 0.780

Table 13: Test RMSLE for depth 2-3 meters. The best model for each scenario is highlighted in bold.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
TOA_15x15_depth_in_2_3	**0.378 ± 0.035**	0.556 ± 0.045	0.398 ± 0.038	0.384 ± 0.046	0.419 ± 0.043	0.589 ± 0.046	0.443 ± 0.047	0.426 ± 0.037	0.432 ± 0.037	0.415 ± 0.047
TOA_9x9_depth_in_2_3	**0.388 ± 0.034**	0.558 ± 0.047	0.404 ± 0.039	0.401 ± 0.043	0.426 ± 0.046	0.599 ± 0.047	0.436 ± 0.048	0.430 ± 0.041	0.431 ± 0.037	0.419 ± 0.049
TOA_5x5_depth_in_2_3	**0.393 ± 0.035**	0.555 ± 0.045	0.408 ± 0.038	0.429 ± 0.038	0.427 ± 0.045	0.575 ± 0.041	0.443 ± 0.058	0.442 ± 0.041	0.440 ± 0.038	0.423 ± 0.038
TOA_3x3_depth_in_2_3	**0.409 ± 0.035**	0.563 ± 0.044	0.419 ± 0.040	0.432 ± 0.041	0.432 ± 0.045	0.592 ± 0.049	0.462 ± 0.058	0.447 ± 0.041	0.450 ± 0.040	0.426 ± 0.041
C2RCC_rhown_5x5_depth_in_2_3	0.424 ± 0.041	0.483 ± 0.039	**0.418 ± 0.040**	0.436 ± 0.044	0.436 ± 0.044	0.513 ± 0.037	0.455 ± 0.046	0.436 ± 0.044	0.430 ± 0.042	0.430 ± 0.042
C2X_rhow_9x9_depth_in_2_3	0.430 ± 0.041	0.494 ± 0.035	0.433 ± 0.038	0.441 ± 0.040	0.456 ± 0.044	0.517 ± 0.041	0.464 ± 0.045	0.460 ± 0.047	**0.422 ± 0.032**	0.453 ± 0.041
C2X-Complex_rhown_9x9_depth_in_2_3	0.426 ± 0.045	0.497 ± 0.046	0.432 ± 0.047	0.455 ± 0.057	0.441 ± 0.045	0.525 ± 0.044	0.466 ± 0.058	0.446 ± 0.048	**0.425 ± 0.046**	0.438 ± 0.045
C2X-Complex_rhow_5x5_depth_in_2_3	**0.430 ± 0.038**	0.510 ± 0.045	0.441 ± 0.044	0.455 ± 0.046	0.456 ± 0.044	0.540 ± 0.049	0.490 ± 0.056	0.455 ± 0.047	0.431 ± 0.039	0.445 ± 0.038
C2X-Complex_rhown_5x5_depth_in_2_3	0.436 ± 0.039	0.509 ± 0.046	0.445 ± 0.042	0.458 ± 0.047	0.460 ± 0.042	0.549 ± 0.045	0.480 ± 0.043	0.459 ± 0.046	**0.433 ± 0.039**	0.452 ± 0.038
C2RCC_rhow_3x3_depth_in_2_3	0.447 ± 0.041	0.508 ± 0.040	0.446 ± 0.042	0.452 ± 0.047	0.471 ± 0.043	0.551 ± 0.044	0.472 ± 0.042	0.470 ± 0.045	**0.440 ± 0.041**	0.468 ± 0.044

Table 14: Train RMSLE for depth 2-3 meters.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
TOA_15x15_depth_in_2_3	0.054 ± 0.026	0.535 ± 0.013	0.233 ± 0.014	0.007 ± 0.001	0.124 ± 0.049	0.517 ± 0.023	0.355 ± 0.065	0.149 ± 0.036	0.363 ± 0.018	0.105 ± 0.058
TOA_9x9_depth_in_2_3	0.048 ± 0.025	0.535 ± 0.014	0.234 ± 0.013	0.007 ± 0.001	0.132 ± 0.051	0.522 ± 0.012	0.354 ± 0.056	0.151 ± 0.034	0.367 ± 0.018	0.108 ± 0.060
TOA_5x5_depth_in_2_3	0.050 ± 0.029	0.532 ± 0.013	0.231 ± 0.013	0.007 ± 0.001	0.125 ± 0.053	0.505 ± 0.009	0.349 ± 0.062	0.160 ± 0.038	0.374 ± 0.019	0.104 ± 0.061
TOA_3x3_depth_in_2_3	0.056 ± 0.032	0.542 ± 0.015	0.240 ± 0.013	0.007 ± 0.001	0.122 ± 0.051	0.519 ± 0.019	0.373 ± 0.070	0.163 ± 0.036	0.383 ± 0.018	0.103 ± 0.065
C2RCC_rhown_5x5_depth_in_2_3	0.123 ± 0.050	0.469 ± 0.008	0.260 ± 0.014	0.007 ± 0.001	0.197 ± 0.039	0.466 ± 0.025	0.400 ± 0.037	0.241 ± 0.050	0.370 ± 0.014	0.170 ± 0.058
C2X_rhow_9x9_depth_in_2_3	0.189 ± 0.060	0.468 ± 0.010	0.281 ± 0.016	0.028 ± 0.005	0.217 ± 0.033	0.489 ± 0.018	0.405 ± 0.057	0.275 ± 0.046	0.354 ± 0.014	0.214 ± 0.066
C2X-Complex_rhown_9x9_depth_in_2_3	0.119 ± 0.072	0.480 ± 0.012	0.269 ± 0.019	0.007 ± 0.001	0.195 ± 0.042	0.489 ± 0.019	0.403 ± 0.054	0.233 ± 0.068	0.347 ± 0.019	0.200 ± 0.065
C2X-Complex_rhow_5x5_depth_in_2_3	0.103 ± 0.061	0.490 ± 0.014	0.266 ± 0.019	0.007 ± 0.001	0.163 ± 0.061	0.505 ± 0.021	0.417 ± 0.060	0.243 ± 0.067	0.344 ± 0.023	0.165 ± 0.085
C2X-Complex_rhown_5x5_depth_in_2_3	0.135 ± 0.076	0.493 ± 0.012	0.280 ± 0.017	0.007 ± 0.001	0.200 ± 0.042	0.502 ± 0.017	0.421 ± 0.052	0.262 ± 0.065	0.357 ± 0.024	0.205 ± 0.069
C2RCC_rhow_3x3_depth_in_2_3	0.129 ± 0.067	0.492 ± 0.009	0.277 ± 0.016	0.007 ± 0.001	0.213 ± 0.042	0.503 ± 0.024	0.403 ± 0.043	0.263 ± 0.049	0.375 ± 0.012	0.210 ± 0.063

Table 15: Test R^2 for depth 2-3 meters. The best model for each scenario is highlighted in bold.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
TOA_15x15_depth_in_2_3	**0.723 ± 0.091**	0.467 ± 0.094	0.700 ± 0.062	0.716 ± 0.095	0.636 ± 0.117	0.410 ± 0.138	0.649 ± 0.096	0.651 ± 0.098	0.558 ± 0.068	0.608 ± 0.134
TOA_9x9_depth_in_2_3	**0.706 ± 0.091**	0.455 ± 0.100	0.695 ± 0.060	0.691 ± 0.090	0.639 ± 0.110	0.375 ± 0.192	0.637 ± 0.127	0.649 ± 0.118	0.562 ± 0.071	0.620 ± 0.120
TOA_5x5_depth_in_2_3	**0.697 ± 0.089**	0.476 ± 0.094	0.686 ± 0.056	0.653 ± 0.082	0.631 ± 0.118	0.443 ± 0.135	0.639 ± 0.110	0.632 ± 0.101	0.550 ± 0.066	0.610 ± 0.113
TOA_3x3_depth_in_2_3	**0.692 ± 0.078**	0.463 ± 0.086	0.675 ± 0.058	0.639 ± 0.082	0.634 ± 0.111	0.397 ± 0.143	0.607 ± 0.145	0.626 ± 0.088	0.526 ± 0.068	0.623 ± 0.111
C2RCC_rhown_5x5_depth_in_2_3	0.663 ± 0.085	0.582 ± 0.091	**0.681 ± 0.075**	0.650 ± 0.099	0.625 ± 0.119	0.526 ± 0.153	0.650 ± 0.097	0.638 ± 0.113	0.660 ± 0.076	0.641 ± 0.099
C2X-Complex_rhown_5x5_depth_in_2_3	0.652 ± 0.087	0.477 ± 0.182	0.641 ± 0.085	0.600 ± 0.120	0.607 ± 0.103	0.320 ± 0.353	0.598 ± 0.115	0.612 ± 0.104	**0.670 ± 0.073**	0.596 ± 0.086
C2X-Complex_rhown_9x9_depth_in_2_3	0.635 ± 0.099	0.528 ± 0.124	0.641 ± 0.089	0.585 ± 0.144	0.598 ± 0.124	0.461 ± 0.177	0.601 ± 0.133	0.599 ± 0.118	**0.665 ± 0.079**	0.591 ± 0.102
C2X-Complex_rhow_5x5_depth_in_2_3	**0.664 ± 0.085**	0.477 ± 0.186	0.645 ± 0.085	0.599 ± 0.117	0.610 ± 0.105	0.390 ± 0.282	0.571 ± 0.143	0.613 ± 0.101	0.658 ± 0.079	0.605 ± 0.097
C2RCC_rhow_3x3_depth_in_2_3	0.634 ± 0.095	0.524 ± 0.086	0.643 ± 0.078	0.633 ± 0.113	0.578 ± 0.116	0.429 ± 0.189	0.625 ± 0.090	0.601 ± 0.103	**0.653 ± 0.081**	0.562 ± 0.100
C2X_rhow_9x9_depth_in_2_3	0.644 ± 0.091	0.590 ± 0.085	**0.646 ± 0.071**	0.637 ± 0.079	0.577 ± 0.106	0.559 ± 0.092	0.628 ± 0.111	0.589 ± 0.097	0.628 ± 0.074	0.569 ± 0.098

Table 16: Train R^2 for depth 2-3 meters.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
TOA_15x15_depth_in_2_3	0.998 ± 0.002	0.544 ± 0.021	0.922 ± 0.016	1.000 ± 0.000	0.985 ± 0.013	0.623 ± 0.050	0.786 ± 0.093	0.960 ± 0.025	0.663 ± 0.046	0.937 ± 0.099
TOA_9x9_depth_in_2_3	0.999 ± 0.001	0.536 ± 0.025	0.924 ± 0.013	1.000 ± 0.000	0.983 ± 0.014	0.636 ± 0.020	0.771 ± 0.111	0.964 ± 0.021	0.666 ± 0.046	0.951 ± 0.087
TOA_5x5_depth_in_2_3	0.998 ± 0.002	0.554 ± 0.023	0.926 ± 0.013	1.000 ± 0.000	0.984 ± 0.014	0.646 ± 0.017	0.805 ± 0.088	0.959 ± 0.026	0.659 ± 0.050	0.937 ± 0.105
TOA_3x3_depth_in_2_3	0.998 ± 0.003	0.534 ± 0.030	0.920 ± 0.014	1.000 ± 0.000	0.984 ± 0.014	0.622 ± 0.037	0.772 ± 0.117	0.960 ± 0.025	0.648 ± 0.050	0.934 ± 0.105
C2RCC_rhown_5x5_depth_in_2_3	0.990 ± 0.009	0.630 ± 0.020	0.910 ± 0.015	1.000 ± 0.000	0.970 ± 0.014	0.668 ± 0.065	0.747 ± 0.064	0.908 ± 0.040	0.742 ± 0.026	0.942 ± 0.088
C2X-Complex_rhown_5x5_depth_in_2_3	0.983 ± 0.019	0.582 ± 0.030	0.895 ± 0.022	1.000 ± 0.000	0.970 ± 0.013	0.604 ± 0.074	0.718 ± 0.076	0.898 ± 0.056	0.769 ± 0.038	0.866 ± 0.151
C2X-Complex_rhown_9x9_depth_in_2_3	0.987 ± 0.021	0.604 ± 0.031	0.903 ± 0.021	1.000 ± 0.000	0.973 ± 0.012	0.596 ± 0.066	0.734 ± 0.083	0.916 ± 0.051	0.773 ± 0.035	0.889 ± 0.136
C2X-Complex_rhow_5x5_depth_in_2_3	0.991 ± 0.013	0.594 ± 0.043	0.905 ± 0.025	1.000 ± 0.000	0.979 ± 0.014	0.579 ± 0.066	0.704 ± 0.113	0.914 ± 0.057	0.790 ± 0.037	0.887 ± 0.150
C2RCC_rhow_3x3_depth_in_2_3	0.987 ± 0.015	0.577 ± 0.020	0.892 ± 0.021	1.000 ± 0.000	0.964 ± 0.015	0.583 ± 0.070	0.736 ± 0.077	0.901 ± 0.038	0.737 ± 0.023	0.864 ± 0.152
C2X_rhow_9x9_depth_in_2_3	0.974 ± 0.017	0.661 ± 0.025	0.897 ± 0.019	1.000 ± 0.000	0.965 ± 0.012	0.635 ± 0.038	0.736 ± 0.091	0.893 ± 0.045	0.773 ± 0.035	0.867 ± 0.150

Table 17: Test RMSE for depth 2-3 meters. The best model for each scenario is highlighted in bold.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
TOA_15x15_depth_in_2_3	**1.829 ± 0.359**	2.550 ± 0.283	1.916 ± 0.288	1.838 ± 0.278	2.093 ± 0.394	2.672 ± 0.315	2.057 ± 0.313	2.057 ± 0.357	2.334 ± 0.325	2.169 ± 0.435
TOA_9x9_depth_in_2_3	**1.885 ± 0.346**	2.579 ± 0.292	1.932 ± 0.273	1.922 ± 0.243	2.083 ± 0.345	2.735 ± 0.380	2.079 ± 0.380	2.054 ± 0.380	2.323 ± 0.327	2.139 ± 0.416
TOA_5x5_depth_in_2_3	**1.914 ± 0.334**	2.527 ± 0.269	1.961 ± 0.264	2.049 ± 0.254	2.107 ± 0.378	2.595 ± 0.304	2.081 ± 0.345	2.114 ± 0.349	2.355 ± 0.313	2.173 ± 0.371
TOA_3x3_depth_in_2_3	**1.933 ± 0.284**	2.562 ± 0.257	1.994 ± 0.250	2.090 ± 0.241	2.097 ± 0.339	2.698 ± 0.310	2.160 ± 0.380	2.130 ± 0.296	2.414 ± 0.307	2.132 ± 0.347
C2RCC_rhown_5x5_depth_in_2_3	2.013 ± 0.265	2.244 ± 0.222	**1.961 ± 0.251**	2.047 ± 0.286	2.114 ± 0.331	2.371 ± 0.305	2.050 ± 0.300	2.081 ± 0.303	2.025 ± 0.234	2.077 ± 0.306
C2X-Complex_rhown_5x5_depth_in_2_3	2.045 ± 0.255	2.495 ± 0.360	2.082 ± 0.253	2.183 ± 0.302	2.170 ± 0.274	2.793 ± 0.587	2.195 ± 0.299	2.158 ± 0.282	**1.996 ± 0.241**	2.210 ± 0.269
C2X-Complex_rhow_5x5_depth_in_2_3	**2.008 ± 0.256**	2.492 ± 0.362	2.068 ± 0.258	2.188 ± 0.303	2.161 ± 0.276	2.665 ± 0.458	2.258 ± 0.356	2.157 ± 0.286	2.030 ± 0.248	2.182 ± 0.322
C2X-Complex_rhown_9x9_depth_in_2_3	2.097 ± 0.304	2.383 ± 0.277	2.081 ± 0.281	2.222 ± 0.344	2.195 ± 0.340	2.535 ± 0.337	2.178 ± 0.350	2.194 ± 0.317	**2.011 ± 0.259**	2.225 ± 0.339
C2RCC_rhow_3x3_depth_in_2_3	2.097 ± 0.293	2.402 ± 0.227	2.079 ± 0.256	2.094 ± 0.312	2.249 ± 0.309	2.612 ± 0.334	2.124 ± 0.260	2.190 ± 0.286	**2.043 ± 0.239**	2.303 ± 0.331
C2X_rhow_9x9_depth_in_2_3	2.072 ± 0.292	2.224 ± 0.230	**2.072 ± 0.237**	2.096 ± 0.255	2.257 ± 0.305	2.306 ± 0.232	2.106 ± 0.285	2.226 ± 0.265	2.124 ± 0.271	2.286 ± 0.332

Table 18: Train RMSE for depth 2-3 meters.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
TOA_15x15_depth_in_2_3	0.129 ± 0.069	2.358 ± 0.067	0.969 ± 0.102	0.008 ± 0.002	0.380 ± 0.185	2.142 ± 0.130	1.580 ± 0.343	0.669 ± 0.202	2.022 ± 0.151	0.608 ± 0.639
TOA_9x9_depth_in_2_3	0.112 ± 0.066	2.379 ± 0.079	0.961 ± 0.088	0.008 ± 0.002	0.417 ± 0.195	2.107 ± 0.062	1.632 ± 0.371	0.644 ± 0.179	2.012 ± 0.150	0.542 ± 0.552
TOA_5x5_depth_in_2_3	0.117 ± 0.077	2.332 ± 0.072	0.949 ± 0.085	0.008 ± 0.002	0.392 ± 0.200	2.078 ± 0.062	1.508 ± 0.317	0.675 ± 0.210	2.034 ± 0.165	0.583 ± 0.651
TOA_3x3_depth_in_2_3	0.131 ± 0.087	2.382 ± 0.087	0.984 ± 0.086	0.008 ± 0.002	0.384 ± 0.199	2.146 ± 0.102	1.626 ± 0.378	0.670 ± 0.196	2.066 ± 0.154	0.602 ± 0.657
C2RCC_rhown_5x5_depth_in_2_3	0.314 ± 0.152	2.109 ± 0.059	1.038 ± 0.084	0.008 ± 0.001	0.577 ± 0.145	1.991 ± 0.194	1.730 ± 0.198	1.024 ± 0.243	1.761 ± 0.092	0.693 ± 0.479
C2X-Complex_rhown_5x5_depth_in_2_3	0.373 ± 0.246	2.239 ± 0.088	1.117 ± 0.117	0.008 ± 0.001	0.580 ± 0.142	2.171 ± 0.211	1.825 ± 0.239	1.061 ± 0.317	1.661 ± 0.142	1.051 ± 0.713
C2X-Complex_rhow_5x5_depth_in_2_3	0.270 ± 0.192	2.206 ± 0.122	1.058 ± 0.129	0.008 ± 0.001	0.463 ± 0.198	2.242 ± 0.184	1.861 ± 0.323	0.958 ± 0.339	1.581 ± 0.134	0.868 ± 0.775
C2X-Complex_rhown_9x9_depth_in_2_3	0.315 ± 0.241	2.181 ± 0.092	1.073 ± 0.113	0.008 ± 0.001	0.550 ± 0.142	2.196 ± 0.185	1.768 ± 0.259	0.950 ± 0.324	1.645 ± 0.128	0.955 ± 0.648
C2RCC_rhow_3x3_depth_in_2_3	0.342 ± 0.216	2.256 ± 0.059	1.135 ± 0.108	0.008 ± 0.001	0.641 ± 0.149	2.231 ± 0.194	1.762 ± 0.248	1.063 ± 0.237	1.777 ± 0.085	1.072 ± 0.687
C2X_rhow_9x9_depth_in_2_3	0.522 ± 0.201	2.017 ± 0.069	1.105 ± 0.107	0.045 ± 0.008	0.640 ± 0.119	2.091 ± 0.117	1.758 ± 0.296	1.107 ± 0.258	1.646 ± 0.128	1.047 ± 0.708

Table 19: Test RMSLE for depth 3-4 meters. The best model for each scenario is highlighted in bold.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
TOA_15x15_depth_in_3_4	**0.392 ± 0.035**	0.539 ± 0.038	0.406 ± 0.036	0.397 ± 0.041	0.437 ± 0.046	0.583 ± 0.046	0.451 ± 0.044	0.439 ± 0.043	0.428 ± 0.034	0.421 ± 0.044
TOA_9x9_depth_in_3_4	**0.396 ± 0.033**	0.533 ± 0.040	0.407 ± 0.036	0.406 ± 0.041	0.438 ± 0.045	0.565 ± 0.049	0.448 ± 0.046	0.443 ± 0.039	0.426 ± 0.035	0.425 ± 0.042
TOA_5x5_depth_in_3_4	**0.408 ± 0.037**	0.529 ± 0.033	0.414 ± 0.034	0.438 ± 0.036	0.447 ± 0.042	0.544 ± 0.043	0.461 ± 0.047	0.461 ± 0.044	0.434 ± 0.033	0.434 ± 0.040
TOA_3x3_depth_in_3_4	**0.424 ± 0.035**	0.532 ± 0.033	0.427 ± 0.033	0.439 ± 0.036	0.468 ± 0.042	0.561 ± 0.046	0.477 ± 0.056	0.470 ± 0.037	0.436 ± 0.034	0.444 ± 0.041
C2X-Complex_rhow_15x15_depth_in_3_4	**0.432 ± 0.044**	0.504 ± 0.039	0.440 ± 0.043	0.448 ± 0.050	0.457 ± 0.050	0.530 ± 0.046	0.485 ± 0.045	0.456 ± 0.046	0.433 ± 0.042	0.443 ± 0.044
TOA_1x1_depth_in_3_4	**0.434 ± 0.036**	0.551 ± 0.033	0.440 ± 0.030	0.456 ± 0.038	0.474 ± 0.038	0.597 ± 0.039	0.484 ± 0.053	0.478 ± 0.037	0.448 ± 0.032	0.452 ± 0.037
C2X-Complex_rhow_9x9_depth_in_3_4	0.447 ± 0.048	0.508 ± 0.041	0.450 ± 0.045	0.464 ± 0.052	0.461 ± 0.050	0.521 ± 0.043	0.495 ± 0.047	0.465 ± 0.048	**0.442 ± 0.044**	0.452 ± 0.044
C2X-Complex_rhown_9x9_depth_in_3_4	0.452 ± 0.050	0.508 ± 0.039	0.453 ± 0.045	0.472 ± 0.050	0.474 ± 0.053	0.523 ± 0.045	0.496 ± 0.054	0.474 ± 0.050	**0.448 ± 0.044**	0.456 ± 0.044
C2X-Complex_rhow_5x5_depth_in_3_4	**0.456 ± 0.044**	0.523 ± 0.043	0.462 ± 0.044	0.473 ± 0.050	0.481 ± 0.049	0.544 ± 0.050	0.503 ± 0.051	0.482 ± 0.048	0.460 ± 0.043	0.467 ± 0.043
C2X-Complex_rhown_5x5_depth_in_3_4	**0.456 ± 0.046**	0.517 ± 0.039	0.464 ± 0.043	0.480 ± 0.051	0.479 ± 0.051	0.539 ± 0.044	0.505 ± 0.054	0.482 ± 0.047	0.460 ± 0.042	0.466 ± 0.040

Table 20: Train RMSLE for depth 3-4 meters.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
TOA_15x15_depth_in_3_4	0.058 ± 0.030	0.511 ± 0.013	0.231 ± 0.015	0.004 ± 0.001	0.135 ± 0.053	0.506 ± 0.039	0.357 ± 0.073	0.174 ± 0.044	0.358 ± 0.019	0.129 ± 0.060
TOA_9x9_depth_in_3_4	0.051 ± 0.028	0.505 ± 0.017	0.231 ± 0.012	0.004 ± 0.001	0.139 ± 0.055	0.497 ± 0.037	0.368 ± 0.056	0.167 ± 0.038	0.362 ± 0.016	0.139 ± 0.050
TOA_5x5_depth_in_3_4	0.065 ± 0.036	0.501 ± 0.015	0.232 ± 0.016	0.004 ± 0.001	0.160 ± 0.053	0.477 ± 0.041	0.380 ± 0.064	0.187 ± 0.059	0.366 ± 0.018	0.131 ± 0.060
TOA_3x3_depth_in_3_4	0.070 ± 0.041	0.506 ± 0.017	0.241 ± 0.017	0.004 ± 0.001	0.157 ± 0.050	0.499 ± 0.046	0.400 ± 0.077	0.198 ± 0.056	0.369 ± 0.016	0.142 ± 0.055
C2X-Complex_rhow_15x15_depth_in_3_4	0.130 ± 0.074	0.472 ± 0.017	0.272 ± 0.016	0.005 ± 0.001	0.183 ± 0.042	0.494 ± 0.015	0.414 ± 0.049	0.262 ± 0.055	0.356 ± 0.012	0.192 ± 0.072
TOA_1x1_depth_in_3_4	0.067 ± 0.036	0.522 ± 0.014	0.250 ± 0.015	0.004 ± 0.001	0.180 ± 0.047	0.522 ± 0.038	0.403 ± 0.065	0.213 ± 0.055	0.373 ± 0.016	0.157 ± 0.059
C2X-Complex_rhow_9x9_depth_in_3_4	0.163 ± 0.081	0.482 ± 0.015	0.285 ± 0.017	0.005 ± 0.001	0.189 ± 0.041	0.497 ± 0.010	0.436 ± 0.050	0.273 ± 0.061	0.362 ± 0.016	0.211 ± 0.058
C2X-Complex_rhown_9x9_depth_in_3_4	0.133 ± 0.083	0.488 ± 0.012	0.288 ± 0.018	0.005 ± 0.001	0.203 ± 0.042	0.496 ± 0.013	0.440 ± 0.057	0.307 ± 0.053	0.365 ± 0.016	0.224 ± 0.068
C2X-Complex_rhow_5x5_depth_in_3_4	0.142 ± 0.082	0.496 ± 0.012	0.289 ± 0.021	0.005 ± 0.001	0.184 ± 0.052	0.506 ± 0.018	0.443 ± 0.061	0.294 ± 0.068	0.375 ± 0.020	0.232 ± 0.071
C2X-Complex_rhown_5x5_depth_in_3_4	0.161 ± 0.080	0.499 ± 0.010	0.299 ± 0.021	0.005 ± 0.001	0.211 ± 0.041	0.511 ± 0.014	0.446 ± 0.068	0.305 ± 0.068	0.385 ± 0.021	0.248 ± 0.058

Table 21: Test R^2 for depth 3-4 meters. The best model for each scenario is highlighted in bold.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
TOA_15x15_depth_in_3_4	**0.598 ± 0.092**	0.379 ± 0.119	0.587 ± 0.083	0.565 ± 0.118	0.488 ± 0.160	0.269 ± 0.168	0.510 ± 0.137	0.505 ± 0.147	0.491 ± 0.100	0.480 ± 0.130
TOA_9x9_depth_in_3_4	**0.587 ± 0.091**	0.377 ± 0.114	0.584 ± 0.086	0.556 ± 0.115	0.505 ± 0.143	0.338 ± 0.155	0.501 ± 0.177	0.521 ± 0.126	0.498 ± 0.097	0.485 ± 0.128
TOA_5x5_depth_in_3_4	0.566 ± 0.093	0.396 ± 0.108	**0.575 ± 0.078**	0.494 ± 0.115	0.490 ± 0.132	0.384 ± 0.146	0.503 ± 0.138	0.477 ± 0.130	0.482 ± 0.097	0.476 ± 0.120
TOA_3x3_depth_in_3_4	0.538 ± 0.118	0.392 ± 0.100	**0.549 ± 0.090**	0.492 ± 0.117	0.459 ± 0.154	0.332 ± 0.156	0.473 ± 0.155	0.465 ± 0.124	0.467 ± 0.105	0.446 ± 0.126
TOA_1x1_depth_in_3_4	0.510 ± 0.118	0.362 ± 0.122	**0.526 ± 0.082**	0.464 ± 0.126	0.424 ± 0.157	0.289 ± 0.147	0.437 ± 0.153	0.413 ± 0.174	0.458 ± 0.103	0.417 ± 0.142
C2X-Complex_rhow_9x9_depth_in_3_4	0.474 ± 0.137	0.408 ± 0.142	0.499 ± 0.109	0.466 ± 0.149	0.429 ± 0.176	0.384 ± 0.181	0.448 ± 0.135	0.436 ± 0.163	**0.524 ± 0.085**	0.439 ± 0.097
C2X-Complex_rhow_15x15_depth_in_3_4	0.507 ± 0.127	0.428 ± 0.123	0.518 ± 0.108	0.492 ± 0.165	0.420 ± 0.196	0.378 ± 0.140	0.469 ± 0.111	0.454 ± 0.146	**0.523 ± 0.091**	0.443 ± 0.165
C2X-Complex_rhown_9x9_depth_in_3_4	0.435 ± 0.152	0.410 ± 0.123	0.474 ± 0.108	0.427 ± 0.149	0.361 ± 0.195	0.340 ± 0.203	0.443 ± 0.125	0.402 ± 0.166	**0.498 ± 0.090**	0.414 ± 0.095
C2X-Complex_rhow_5x5_depth_in_3_4	0.477 ± 0.137	0.381 ± 0.170	**0.483 ± 0.118**	0.447 ± 0.164	0.415 ± 0.180	0.345 ± 0.232	0.444 ± 0.141	0.412 ± 0.177	0.466 ± 0.110	0.404 ± 0.119
C2X-Complex_rhown_5x5_depth_in_3_4	0.461 ± 0.172	0.394 ± 0.155	**0.475 ± 0.121**	0.432 ± 0.180	0.408 ± 0.190	0.335 ± 0.211	0.430 ± 0.162	0.409 ± 0.160	0.455 ± 0.100	0.406 ± 0.099

Table 22: Train R^2 for depth 3-4 meters.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
TOA_15x15_depth_in_3_4	0.997 ± 0.003	0.499 ± 0.026	0.899 ± 0.022	1.000 ± 0.000	0.972 ± 0.023	0.545 ± 0.105	0.729 ± 0.143	0.919 ± 0.045	0.664 ± 0.043	0.879 ± 0.138
TOA_9x9_depth_in_3_4	0.998 ± 0.002	0.494 ± 0.039	0.899 ± 0.019	1.000 ± 0.000	0.972 ± 0.023	0.576 ± 0.103	0.706 ± 0.128	0.937 ± 0.035	0.666 ± 0.039	0.874 ± 0.138
TOA_5x5_depth_in_3_4	0.996 ± 0.004	0.511 ± 0.035	0.898 ± 0.022	1.000 ± 0.000	0.964 ± 0.024	0.595 ± 0.095	0.710 ± 0.134	0.911 ± 0.069	0.662 ± 0.041	0.879 ± 0.136
TOA_3x3_depth_in_3_4	0.996 ± 0.007	0.497 ± 0.040	0.888 ± 0.025	1.000 ± 0.000	0.962 ± 0.026	0.559 ± 0.113	0.679 ± 0.153	0.911 ± 0.061	0.659 ± 0.038	0.859 ± 0.141
TOA_1x1_depth_in_3_4	0.996 ± 0.004	0.482 ± 0.038	0.880 ± 0.024	1.000 ± 0.000	0.951 ± 0.024	0.554 ± 0.109	0.665 ± 0.133	0.881 ± 0.069	0.660 ± 0.039	0.843 ± 0.146
C2X-Complex_rhow_9x9_depth_in_3_4	0.968 ± 0.034	0.513 ± 0.035	0.849 ± 0.027	1.000 ± 0.000	0.961 ± 0.018	0.491 ± 0.022	0.596 ± 0.096	0.834 ± 0.077	0.669 ± 0.038	0.817 ± 0.164
C2X-Complex_rhow_15x15_depth_in_3_4	0.980 ± 0.027	0.529 ± 0.038	0.860 ± 0.026	1.000 ± 0.000	0.964 ± 0.019	0.498 ± 0.046	0.637 ± 0.099	0.848 ± 0.073	0.662 ± 0.031	0.830 ± 0.176
C2X-Complex_rhown_9x9_depth_in_3_4	0.977 ± 0.032	0.493 ± 0.032	0.836 ± 0.031	1.000 ± 0.000	0.956 ± 0.017	0.477 ± 0.045	0.587 ± 0.112	0.786 ± 0.080	0.649 ± 0.043	0.755 ± 0.181
C2X-Complex_rhow_5x5_depth_in_3_4	0.974 ± 0.030	0.485 ± 0.040	0.839 ± 0.035	1.000 ± 0.000	0.963 ± 0.020	0.495 ± 0.054	0.588 ± 0.136	0.816 ± 0.095	0.637 ± 0.055	0.730 ± 0.180
C2X-Complex_rhown_5x5_depth_in_3_4	0.968 ± 0.030	0.477 ± 0.028	0.824 ± 0.033	1.000 ± 0.000	0.954 ± 0.017	0.470 ± 0.042	0.594 ± 0.128	0.791 ± 0.097	0.611 ± 0.053	0.690 ± 0.157

Table 23: Test RMSE for depth 3-4 meters. The best model for each scenario is highlighted in bold.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
TOA_15x15_depth_in_3_4	**1.889 ± 0.342**	2.344 ± 0.322	1.918 ± 0.331	1.957 ± 0.369	2.103 ± 0.336	2.529 ± 0.330	2.069 ± 0.333	2.083 ± 0.378	2.124 ± 0.325	2.139 ± 0.358
TOA_9x9_depth_in_3_4	**1.915 ± 0.330**	2.348 ± 0.323	1.923 ± 0.333	1.979 ± 0.363	2.077 ± 0.343	2.408 ± 0.333	2.079 ± 0.367	2.054 ± 0.360	2.110 ± 0.319	2.137 ± 0.377
TOA_5x5_depth_in_3_4	1.963 ± 0.335	2.313 ± 0.311	**1.946 ± 0.321**	2.115 ± 0.350	2.111 ± 0.325	2.321 ± 0.308	2.085 ± 0.339	2.151 ± 0.377	2.141 ± 0.293	2.155 ± 0.367
TOA_3x3_depth_in_3_4	2.020 ± 0.349	2.323 ± 0.312	**2.001 ± 0.323**	2.118 ± 0.345	2.171 ± 0.332	2.418 ± 0.322	2.146 ± 0.378	2.173 ± 0.345	2.167 ± 0.286	2.210 ± 0.344
TOA_1x1_depth_in_3_4	2.078 ± 0.348	2.375 ± 0.318	**2.053 ± 0.310**	2.175 ± 0.366	2.238 ± 0.300	2.498 ± 0.303	2.223 ± 0.366	2.266 ± 0.361	2.186 ± 0.281	2.267 ± 0.366
C2X-Complex_rhow_9x9_depth_in_3_4	2.149 ± 0.350	2.280 ± 0.323	2.105 ± 0.331	2.159 ± 0.355	2.233 ± 0.389	2.320 ± 0.346	2.197 ± 0.300	2.222 ± 0.353	**2.055 ± 0.305**	2.237 ± 0.363
C2X-Complex_rhow_15x15_depth_in_3_4	2.087 ± 0.328	2.255 ± 0.330	2.074 ± 0.343	2.108 ± 0.348	2.251 ± 0.387	2.348 ± 0.343	2.172 ± 0.335	2.200 ± 0.378	**2.063 ± 0.316**	2.221 ± 0.393
C2X-Complex_rhown_9x9_depth_in_3_4	2.223 ± 0.355	2.280 ± 0.307	2.158 ± 0.332	2.237 ± 0.336	2.357 ± 0.382	2.399 ± 0.363	2.216 ± 0.340	2.289 ± 0.367	**2.109 ± 0.304**	2.289 ± 0.371
C2X-Complex_rhow_5x5_depth_in_3_4	2.138 ± 0.317	2.329 ± 0.339	**2.135 ± 0.321**	2.194 ± 0.348	2.253 ± 0.368	2.382 ± 0.375	2.201 ± 0.289	2.261 ± 0.355	2.171 ± 0.312	2.300 ± 0.358
C2X-Complex_rhown_5x5_depth_in_3_4	2.161 ± 0.348	2.305 ± 0.336	**2.151 ± 0.325**	2.217 ± 0.341	2.264 ± 0.377	2.408 ± 0.374	2.229 ± 0.329	2.272 ± 0.353	2.199 ± 0.319	2.301 ± 0.357

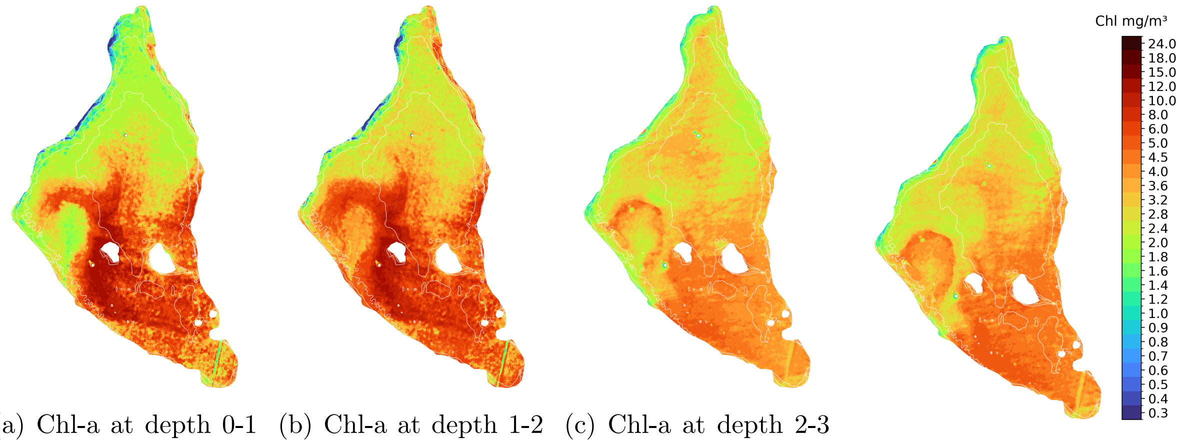
Table 24: Train RMSE for depth 3-4 meters.

Model	CAT	ELN	ENS	KNN	LBM	LR	MLP	RF	SVR	XGB
TOA_15x15_depth_in_3_4	**1.889 ± 0.342**	1.918 ± 0.331	1.957 ± 0.369	2.103 ± 0.336	2.069 ± 0.333	2.124 ± 0.325	2.139 ± 0.358			
TOA_9x9_depth_in_3_4	**1.915 ± 0.330**	1.923 ± 0.333	1.979 ± 0.363	2.077 ± 0.343	2.079 ± 0.367	2.110 ± 0.319	2.137 ± 0.377			
TOA_5x5_depth_in_3_4	1.963 ± 0.335	**1.946 ± 0.321**	2.115 ± 0.350	2.111 ± 0.325	2.085 ± 0.339	2.141 ± 0.293	2.155 ± 0.367			
TOA_3x3_depth_in_3_4	2.020 ± 0.349	**2.001 ± 0.323**	2.118 ± 0.345	2.171 ± 0.332	2.146 ± 0.378	2.167 ± 0.286	2.210 ± 0.344			
TOA_1x1_depth_in_3_4	2.078 ± 0.348	**2.053 ± 0.310**	2.175 ± 0.366	2.238 ± 0.300	2.223 ± 0.366	2.186 ± 0.281	2.267 ± 0.366			
C2X-Complex_rhow_9x9_depth_in_3_4	2.149 ± 0.350	2.105 ± 0.331	2.159 ± 0.355	2.233 ± 0.389	2.197 ± 0.300	**2.055 ± 0.305**	2.237 ± 0.363			
C2X-Complex_rhow_15x15_depth_in_3_4	2.087 ± 0.328	2.074 ± 0.343	2.108 ± 0.348	2.251 ± 0.387	2.172 ± 0.335	**2.063 ± 0.316**	2.221 ± 0.393			
C2X-Complex_rhown_9x9_depth_in_3_4	2.223 ± 0.355	2.158 ± 0.332	2.237 ± 0.336	2.357 ± 0.382	2.216 ± 0.340	**2.109 ± 0.304**	2.289 ± 0.371			
C2X-Complex_rhow_5x5_depth_in_3_4	2.138 ± 0.317	**2.135 ± 0.321**	2.194 ± 0.348	2.253 ± 0.368	2.201 ± 0.289	2.171 ± 0.312	2.300 ± 0.358			
C2X-Complex_rhown_5x5_depth_in_3_4	2.161 ± 0.348	**2.151 ± 0.325**	2.217 ± 0.341	2.264 ± 0.377	2.229 ± 0.329	2.199 ± 0.319	2.301 ± 0.357			

Additional Figures

Complementary to the maps generated for July 14, 2022, and July 28, 2025, additional maps were created.

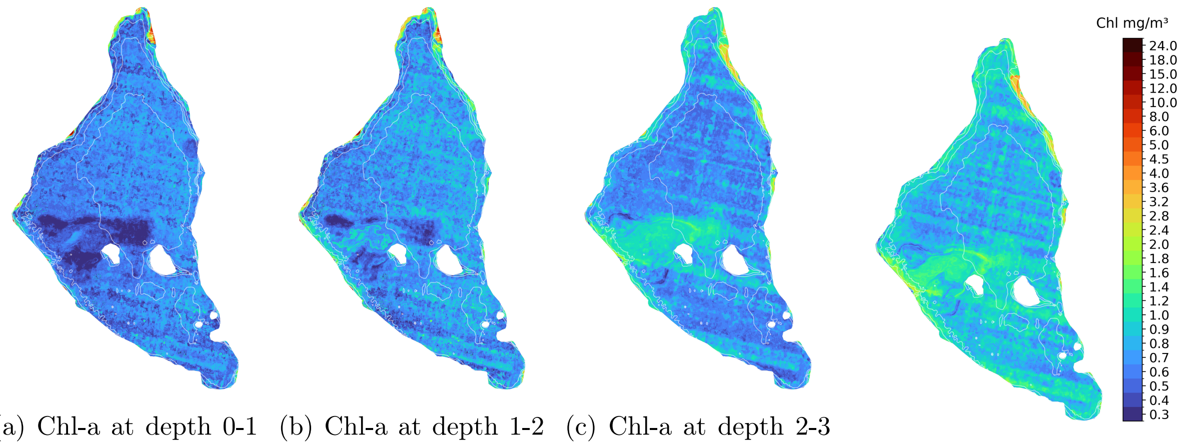
The maps for August 13, 2021, in Figure 1, correspond to a short period in which chlorophyll-a concentrations peaked, with values close to those reached during the 2016 eutrophication crisis.



(a) Chl-a at depth 0-1 meters.
(b) Chl-a at depth 1-2 meters.
(c) Chl-a at depth 2-3 meters.
(d) Chl-a at depth 3-4 meters.

Figure 1: Chl-a concentrations predicted for 2021-08-13

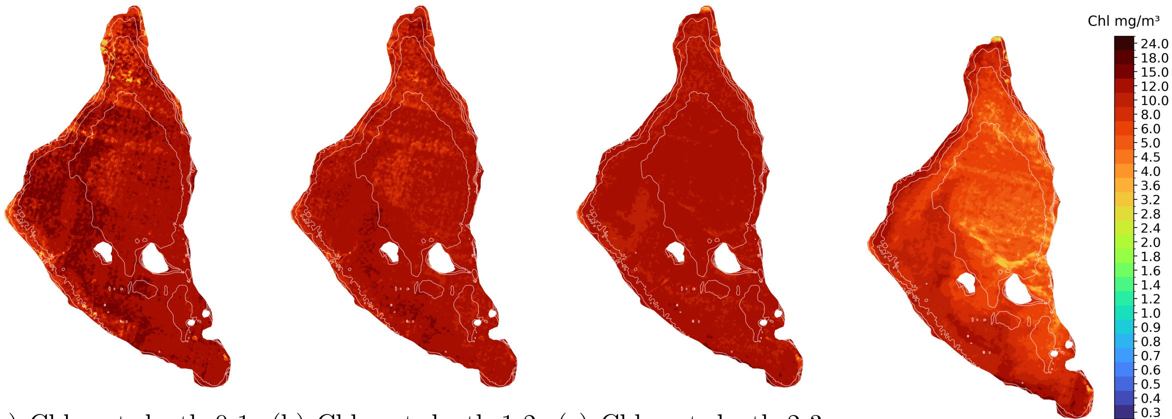
Figure 2 represents Chl-a concentrations for February 4, 2024. This date was represented to visualize normal behavior: low Chl-a values at surface which increase slightly with depth.



(a) Chl-a at depth 0-1 meters.
(b) Chl-a at depth 1-2 meters.
(c) Chl-a at depth 2-3 meters.
(d) Chl-a at depth 3-4 meters.

Figure 2: Chl-a concentrations predicted for 2024-02-04

The last set of maps, in Figure 3, corresponds to September 8, 2016, during the eutrophication crisis. The maps show Chl-a concentrations peaking and in some depths, they are even close to saturate the color scale.



(a) Chl-a at depth 0-1 meters.

(b) Chl-a at depth 1-2 meters.

(c) Chl-a at depth 2-3 meters.

(d) Chl-a at depth 3-4 meters.

Figure 3: Chl-a concentrations predicted for 2016-09-08